HOMEWORK 2.4 · UNIT 2 · NC.8.EE.1

Advanced Exponent Rule Problems

Targets: EE.1-2, EE.1-4, EE.1-3, EE.1-6, EE.1-7, EE.1-5, EE.1-9

Name _____ Date _____ Block _____

Remember: Simplify one step at a time: **powers first**, then **multiply**, then **divide**. Numbers with numbers, letters with letters. End with **positive exponents**. Example: $\frac{(2x^3)^2}{4x^8} = \frac{4x^6}{4x^8} = x^{-2} = \frac{1}{x^2}$.

SCORE 3.0 Approaching · the basics

Targets: EE.1-2, EE.1-4, EE.1-3

1 Simplify. $(3x^4y^2)^2$ EE.1-2

2 Simplify. $(4a^{-2})(3a^5)$ EE.1-4

3 Simplify. Use positive exponents. EE.1-4,EE.1-3

$$\frac{20m^3}{4m^5}$$

4 Simplify. $\frac{-6x^5y^0}{-2x^{-2}}$ EE.1-4

Ⓐ $3x^7$ Ⓑ $3x^3$ Ⓒ $-3x^7$ Ⓓ $4x^7$

5 $(x^2)^5 = x^7$ REVIEW 2.3 EE.1-2

True False

SCORE 4.0 Proficient · on target

Targets: EE.1-6, EE.1-7, EE.1-5

6 Simplify. Use positive exponents. EE.1-6

$(3a^2b^{-1})^{-2}$

Hint: Raise each factor to the −2 power.

7 **Tess** made a mistake. Circle it. Explain it. Fix it. REVIEW 2.2 EE.1-7

TESS'S WORK

$(2x^3)(4x^5) = 6x^8$

8 Simplify. $(4x^2y)^2 \cdot 3xy^3$ EE.1-5

SCORE 5.0 Mastery · go deeper

Targets: EE.1-9

9 Simplify $\frac{(2x^3)^4}{8x^7}$. Justify each step. EE.1-9

First _____ . Next _____ . The answer is _____ .

10 Find the missing exponent: $(y^{\square})^4 \cdot y^2 = y^{14}$. Explain. EE.1-9

_____ × 4 + 2 = 14, so the missing exponent is _____ .